

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2018
Bachelor in Computer Application (BCA)
Course Title: MIS and E-Business

Group B

2. Explain C2C and B2B e-commerce models with example.
3. What is M-Commerce? Explain its benefits for organization, individual and society.
4. Define wireless communication. Differentiate between UMTS and LTE wireless data communication technologies.
5. Discuss how do enterprise application are used in organization.
6. What is digital signature? Explain the requirements of digital signature in secure transaction. How does digit signature work?
7. Explain cryptocurrency with its legal provisions in Nepal. How is it different from virtual currency?
8. What is Search Engine Optimization (SEO)? Explain the importance of SEO for the growth of a website.

Group C

9. Discuss the security requirements of Internet and E-commerce applications and how these requirements are fulfilled by various hardware and software systems? Explain in detail with examples.
10. "The use of electronic payment systems has significantly increased in Nepal in recent years". In this context, explain the trend of cashless transaction in Nepal along with the benefits and downsides of digital payment systems and cashless economy.
11. How can you evaluate whether the proposed e-commerce project is feasible or not. What can be the different funding techniques for a fresh graduate to setup a new online business? Explain

Tribhuvan University
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2018
Bachelor in Computer Application (BCA)
Course Title: Computer Networking
Code No: CACS 303
Semester: V

Group B

2. Explain TCP/IP Reference model.
3. "Slotted ALOHA improves the performance of system over pure ALOHA". Justify.
4. Differentiate between Reliable & Unreliable connection with their application.
5. What is DNS? How it works?
6. What is cryptography? Explain the principles of cryptography.
7. What is TCP? Describe TCP segment header.
8. Write short note on (Any Two):
 - a) Channelization
 - b) Fragmentation
 - c) VPN

Group C

9. Classify and calculate the total number of Networks and Hosts per network in IPV4.
10. Explain the IPV4 Header format in detail.
11. What is switching? Describe the types of switching with their advantages and disadvantages.

Bachelor in Computer Application (BCA)
Course Title: Computer Graphics and Animation
Code No: CACS 305
Semester: V
2018

Group B

2. What do you mean by area filling? Explain boundary fill algorithm.
3. Define scaling transformation? Prove that two successive scaling are multiplicative.
- What is symmetric property of circle? Digitize a circle whose center is at origin and radius is 9.
- Define window and view port. Explain 2D viewing transformation pipeline.
6. Reflect a prism A(0,0,0), B(1,1,0), C(1,2,2) and (0,2,0) about yz-plane which has been rotated previously with +90 degree about y-axis.
7. What do you mean by visible surface detection? Explain Z buffer algorithm for visible surface detection.
8. What is animation? Explain the steps to create animation sequence.

Group C

9. Define refresh rate and persistence of monitor. Explain architecture of random scan display system with block diagram.
10. How can you draw line using Bresenham's line drawing algorithm? Trace it to rasterize the line with end points P (1, 4) and Q (7,8).
11. Explain Cohen Sutherland line clipping algorithm. Let ABCD be the rectangular window with A(20,20). B(90,20), C(90,70) and D(20,70). Use Cohen Sutherland algorithm to find the region codes and clip the lines with end point P(10,30) and Q(80,80)

Bachelor in Computer Application (BCA)
Course Title: Introduction to Management
Code No: CAMG 304
Semester: V
2018

Group B

11. Define management? What kinds of skills required for successful manager?
12. Describe the major characteristics of contingency theory. Why there is "no one best way" to manage
13. What is the planning? Explain the different level of planning.
14. Define organizational structure. What are the differences between centralization and decentralization?
15. Define communication and explain the various steps in the communication process.
16. What is leadership? Explain the different style of leadership.
17. What are the major technique of control? Explain.

Group C

18. Decision making is very important managerial function. With the statement; discuss the meaning and process of decision making. Also explain the types of decision making.
19. Quality improvement has no barriers and limitations. On the light of this statement, explain the concept of Total Quality Management and explain different tools of TQM.
20. Explain the concept of change in organization. Why do employees resist change in organization? And suggest various measures to overcome such resistivity to change.

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2019
Bachelor in Computer Application (BCA)
Course Title: MIS and E-Business

Group B [6x5=30]

Attempts any SIX questions.

2. How properties like ubiquity, richness and information density make e-commerce better than traditional commerce?
3. Differentiate B2C, B2B and C2C e-commerce with examples.
4. What is the role of information superhighway in e-commerce? List the components of information superhighway. [2+3]
5. What is international information system? Describe the concepts of outsourcing and offshoring in the system. [1+4]
6. Define availability, authentication and authorization. How they can be enforced in e-commerce security? [3+2]
7. What is e-checking? Describe its working mechanism. [2+3]
8. Describe the website design criteria that one should show while designing an e-commerce website.

Group C

Attempts any TWO questions. [2x10=20]

9. What do you mean by Wireless Application Protocol (WAP)? How it works? Discuss the layered architecture of WAP. [2+2+6]
10. How important web contents are for e-commerce systems? Discuss their types. Justify, with examples, how web contents can be used to promote cross-selling, up-selling and promotions in e-commerce. [2+2+6]
11. Discuss in detail the security defends strategies that you can implement while securing e-commerce systems from security attacks and threats.

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Group B [6x5=30]

Attempts any SIX questions

2. Define protocols. Explain WWW and HTTP protocol. [1+2+2]
3. Define transmission impairment. Explain the causes of impairments. [1+4]
4. Define HDLC. Explain the HDLC frame formats. [1+4]
5. Define IP Address. Specify IPv4 address classes with their address ranges. [1+4]
6. Define Subnetting. Suppose you are given network address: 192.168.10.0 and subnet mask: 255.255.255.240 then calculate total number of subnets and numbers of hosts per subnet. [1+2+2]
7. Draw a User Datagram format. Explain UPD operations. [2+3]
8. Write short notes on (Any Two): [2.5+2.5]
 - a) DNS
 - b) Public Key Cryptography
 - c) VPN

Group C

Attempt any TWO questions. [2x10=20]

9. Critically analyze the OSI reference model. [10]
10. Explain the random-access protocols under the multiple access taxonomy. [10]
11. Explain the IPv4 Header format in detail. [10]

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2020
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Group B [6x5=30]

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3. Differentiate B2C, B2B and C2C e-commerce with examples.
4. What is the role of information superhighway in e-commerce? List the components of information superhighway. [2+3]
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11. Discuss in detail the security defends strategies that you can implement while securing e-commerce systems from security attacks and threats.

Tribhuvan University
2020
Bachelor in computer Application(BCA)
Course Title: Introduction to management
Semester 5th

Group B

2. What is management? Explain the functions of management.
3. Define scientific management. Explain the major contributions made by F.W. Taylor.
4. Describe the formal and informal group with suitable example.
5. What is planning? Discuss the steps involved in planning process.
6. What are the barriers of effective communication in any organizations? Explain.
7. Why is motivation important to achieve organizational goals?
8. Define delegation of authority. Explain major features of delegation of authority.

Group C

Attempts any TWO questions.

9. Planning is head and controlling is tail. Justify.
10. Define leadership. Explain different style of leadership with suitable example.
11. What are managerial skills? What skills are essential for a successful manager.

Bachelor in Computer Applications
Course Title: DotNet Technolouy
Code No: CAC8302
Semester: V
2021

Group B

2. What do you mean C# is type safety programming language? Explain any four applied technologies in Dot Net framework.
3. List any five contextual keywords in C#? Write a C# program to initialize and display jagged array elements with sum of each row. [1+4]
4. What is string interpolation? How passing argument by value is differing from passing argument by reference? Explain with program. [1 4]
5. What are uses of base keyword in C#? Write a CW program to illustrate the concept of abstract class and abstract method. [1-2--+2]
6. What is virtual method? Write a C# program to illustrate multi-level inheritance with virtual method. [1+4]
7. What is difference between indexer and properties? Explain the concept of generic class in C# with suitable program. [2+3]
8. Write a program to create form for calculating simple interest in one ASP.NET page and display the simple interest in another page of ASP.NET. [5]

Group C

9. a) What is SystemException? Write a C# program that will read balance and withdraw amount from keyboard and display the remaining balance on screen if balance is greater than withdraw amount otherwise throw an ApplicationException with appropriate message. [1+4]
b) List any five LINQ standard operator? Write a C# program to compute aggregate salary of 5 employee and then display employee record in descending order with respect to employee salary using LINQ. | 1+4]
10. What are different type of delegate in C#? Write a C# program to create a class Time which represents time, The class should have three fields for hours, minutes and seconds. It should have constructor to initialize hours, minutes and seconds and method displayTime() to print current time. Overload following operators:
 - a) + (add two time objects on 24 hours clock)
 - b) < (compare two time objects)

11. What is difference between executeScalur() and executeReader() method ?
Write a C# program to connect database Bank and insert 5 customer record (Account_no, Name, Address, Balance) in Customer table and Display the customer record whose balance is greater than 5000.

Batch: 2019
Bachelor in Computer Applications
Course Title: Introduction to Management
Code No: CAMG304
Semester: V

Group B

2. Define management. Describe the functions of management.
3. What is planning? Discuss the process of planning.
4. What are the condition of decision making? Explain.
5. Discuss the steps of controlling.
6. What are the main leadership styles? Explain.
7. Describe the major important techniques that can be used for motivating employees.
8. Define communication? Explain the types communication.

Group C

- 9 Explain contingency theory of management. What are its contributions and limitations?
- 10, "Organizing structuring is the process of designing job, grouping jobs into manageable units, and establishing patterns of authority among jobs and groups of jobs". In line of this statement state the steps involved in the process of structuring an organization. Also explain the principle of organizing.
11. Change can be introduced successfully only when there are conducive factors in existence in the system. In the highlight of the statement discuss the paradigm shifts and areas of organizational change. Also explain the factor overcoming resistant to organizational change.

Bachelor in Computer Applications
Course Title: MIS and e-Business
Code No: CACS301
Semester: V
2021

Group B

2. What has Location-based Commerce done to help online business? Explain its main idea, components, and the applications. [5]
- 3 Define wireless communication. What are the features of Wi-Fi that have made it so popular in the present? [1+4]
4. How do systems for managing relationships with customer's help businesses get closer to their customers?
5. Differentiate between virus, worm and Trojan horse. [5]
6. What is encryption and decryption? Explain the symmetric key encryption and asymmetric key encryption in detail. [1+4]
7. What is virtual currency? Explain the different types of virtual currencies popular in the market.
8. What are the different ways an e-commerce website can be promoted so that it has a strong presence in the market? Explain. [5]

Group C

9. Define e-Commerce. Explain its significance in today's business world. Explain different types of e-commerce models with examples. [2+3+5]
10. "In the digital world we live in now, information systems like Transaction Processing System, Management Information System, Decision Support System, and Executive Support System are a must for any organization." Explain why the statement is true, including what these information systems are and how they help the organization. [10]
11. As more organizations use IT systems, ransomware attacks have become a big problem, especially during and after the Covid 19 pandemic. In this scenario, describe ransomware in detail, including the effects of attacks, different types of ransomware, ways to protect against them, and what to do when they happen. [10]

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2021
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Code No: CACS303
Semester: V

Group B

2. What is digital signature and how it works?
3. Why do we need DNS when we can directly use an IP address? Explain the process to resolve the domain www.tribhuvan-university.edu.np ? [2+3]
4. What do you mean by IP datagram fragmentation?
5. Explain the mechanism of bit and byte stuffing with suitable example.
6. Explain the causes of transmission impairment in data communication.
7. What is flow control? Explain Go-Back-N ARQ with suitable diagram. [1+4]
8. Explain in detail about the structure of IPv6 address. Differentiate between IPv4 and IPv6 address. [3+2]

Group C

9. Explain how slotted Aloha improves the performance of system over pure Aloha. A bit stream of 100101010 is transmitted using CRC method. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted and if fourth bit from LSB is altered, show the error detection in receiver side. [2+8]
10. Why do we need layered protocol architecture? Discuss each layer of TCP/IP architecture along with function of each layer. Write the protocols used in each layer of TCP/IP model. [2+4+4]
11. Why do we need subnetting? What is the new subnet mask? Write subnet ID and broadcast address of each subnet if you divide a class C network (192.6.3.0-192.6.3.255) into 4 different subnets?